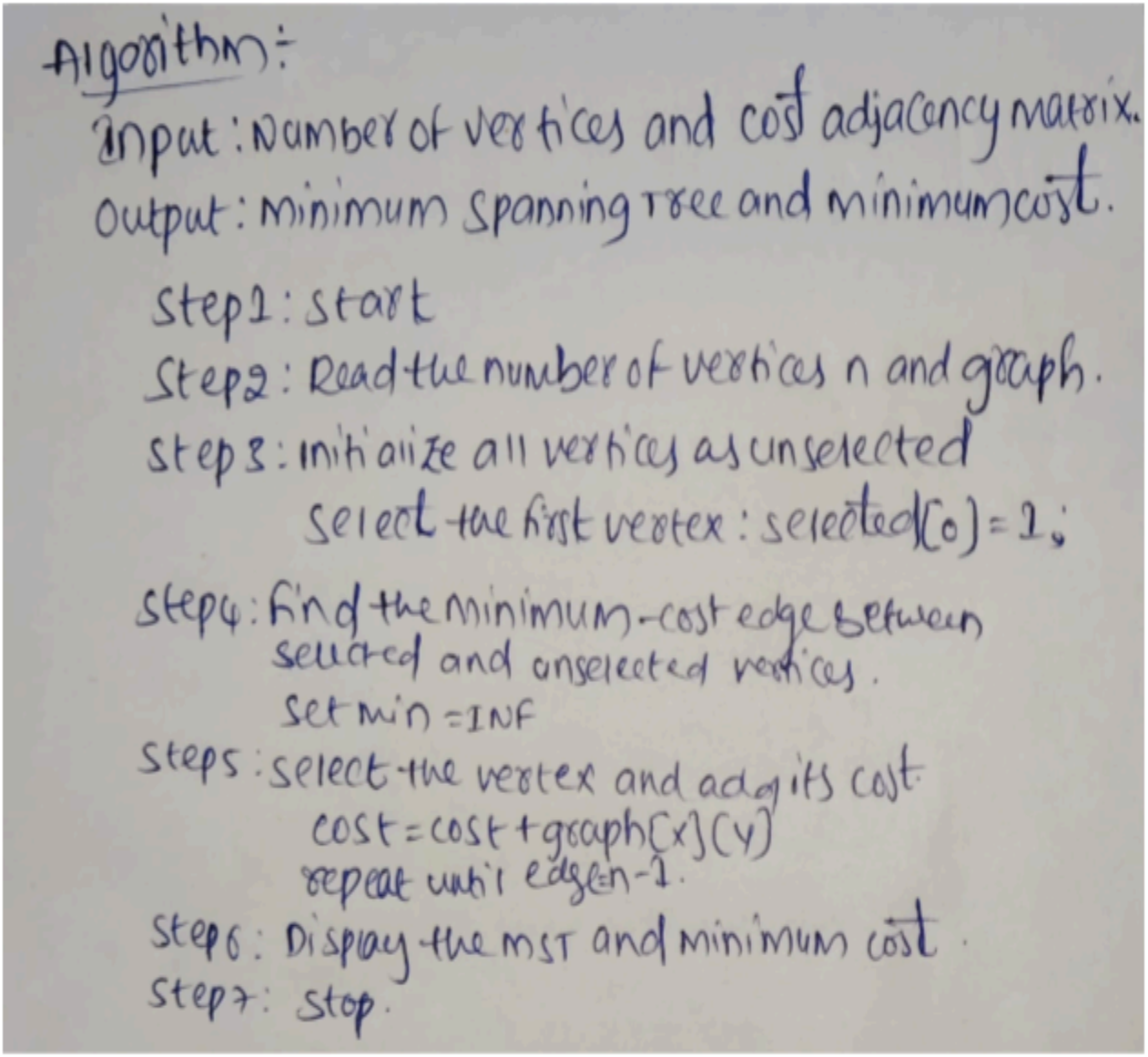
practical-9

Aim: To Implement prim's algorithm

prim's algorithm

Algorithm:



Program:

```
#include <iostream>
using namespace std;
int main()
{
   int graph[MAX][MAX];
   int selected[MAX];
   int n, i, j;
   int edge = 0;
   int x, y;
   int min, cost = 0;

   cout << "Enter number of vertices: ";
   cin >> n;
   cout << "Enter Cost Adjacency Matrix:\n";


   for(i = 0; i < n; i++)


   {
      for(j = 0; j < n; j++)
      {
         cin >> graph[i][j];

         if(graph[i][j] == 0)
            graph[i][j] = INF;
      }
   }

   // Initially select first vertex
   for(i = 0; i < n; i++)
      selected[i] = 0;

   selected[0] = 1;
   cout << "\nEdges in Minimum Spanning Tree:\n";
   while(edge < n - 1)
   {
      min = INF;
```

```
        for(i = 0; i < n; i++)
        {
            if(selected[i])
            {
                for(j = 0; j < n; j++)
                {
                    if(!selected[j] && graph[i][j] < min)
                    {
                        min = graph[i][j];
                        x = i;
                        y = j;
                    }
                }
            }
        }

        cout << x << " --> " << y << " Cost = " << graph[x][y] << endl;

        cost += graph[x][y];
        selected[y] = 1;
        edge++;
    }

    cout << "\nMinimum Cost = " << cost << endl;

    return 0;
}
```

Output:

```
Output

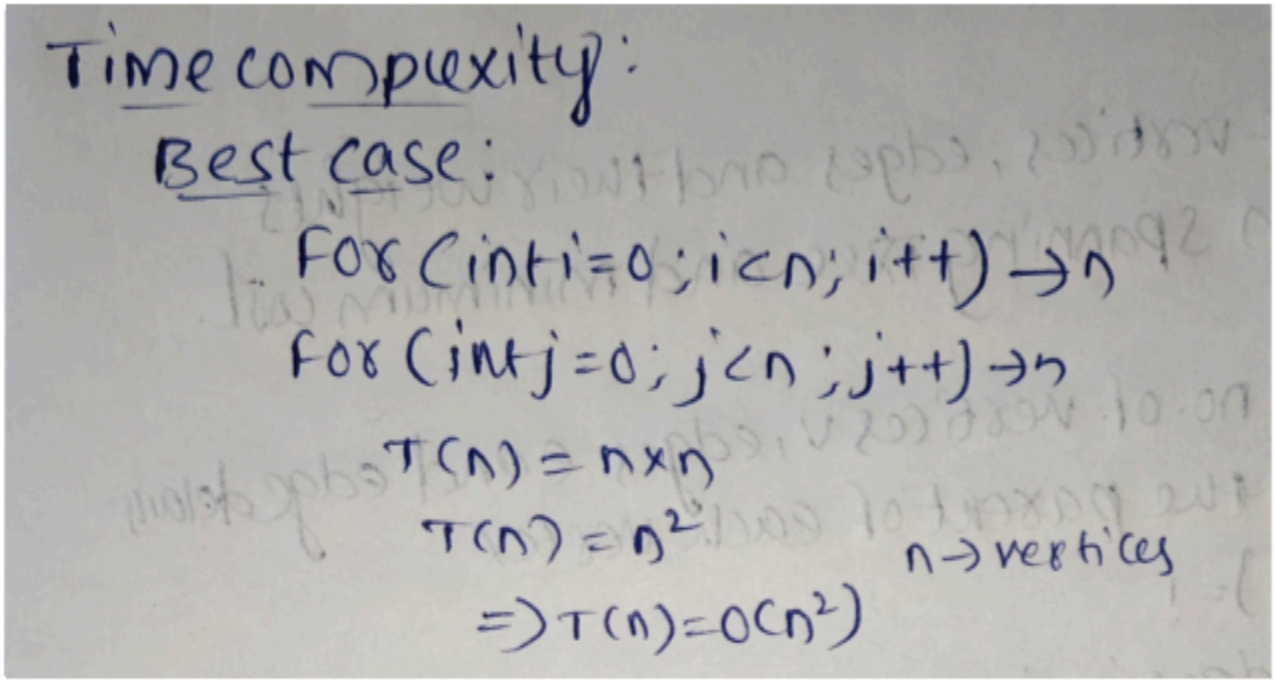
Enter number of vertices: 6
Enter Cost Adjacency Matrix:
0 2 3 0 0 0
2 0 1 4 0 0
3 1 0 5 6 0
0 4 5 0 2 3
0 0 6 2 0 1
0 0 0 3 1 0

Edges in Minimum Spanning Tree:
0 --> 1   Cost = 2
1 --> 2   Cost = 1
1 --> 3   Cost = 4
3 --> 4   Cost = 2
4 --> 5   Cost = 1

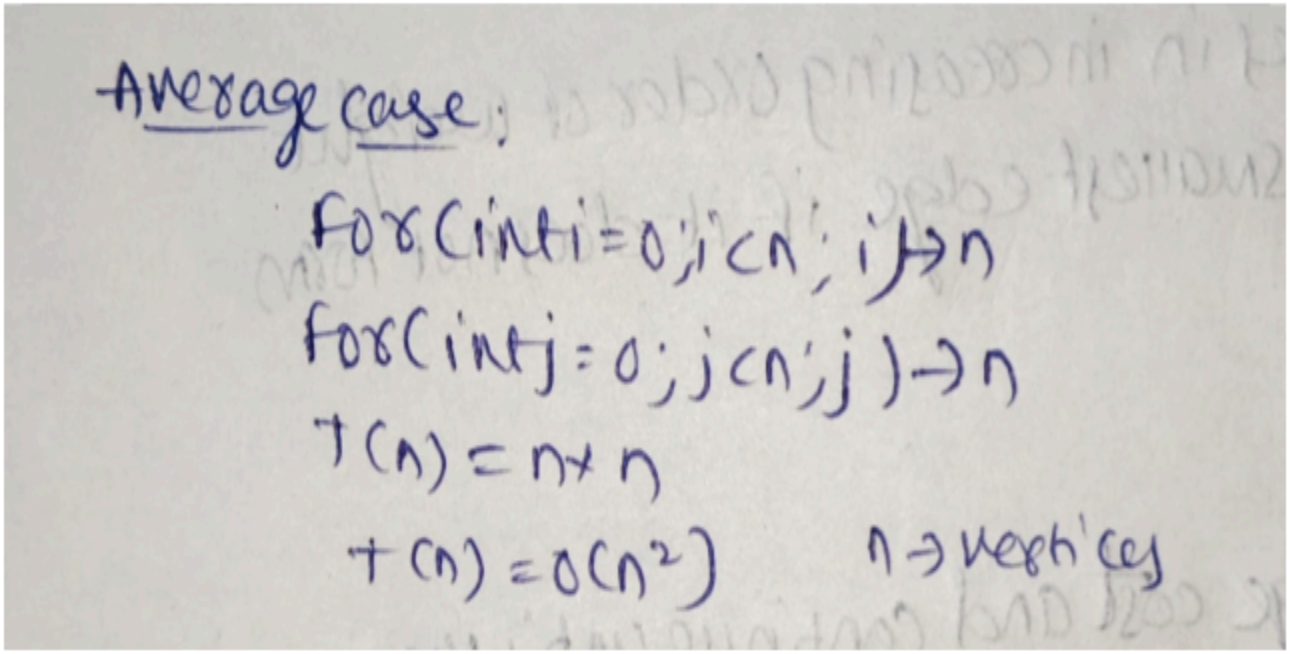
Minimum Cost = 10
```

Time Complexity:

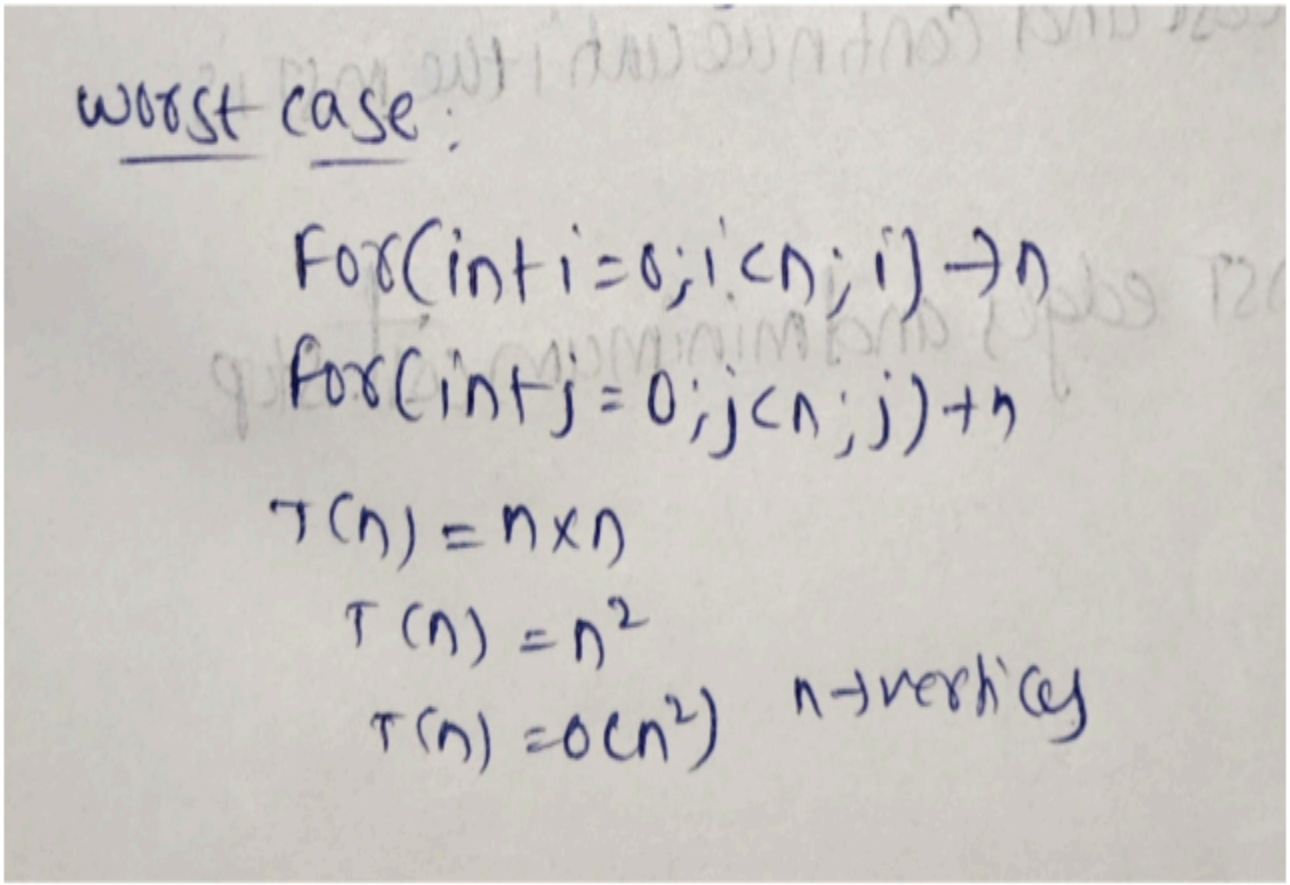
Best Case:



Average Case:



Worst Case:



Conclusion:

The implementation of **Prim's Algorithm** helps us understand how a **Minimum Spanning Tree (MST)** is constructed for a weighted graph. It repeatedly selects the minimum-cost edge connecting a selected vertex to an unselected vertex. For the given adjacency matrix implementation, Prim's Algorithm has a time complexity of $\mathbf{O(n^2)}$ in the best, average, and worst cases. Therefore, Prim's Algorithm is a useful technique for finding the minimum-cost spanning tree of a connected graph.